BEING FINITE IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's own grammatical slip 'An finite morphism' occurs in the source and is preserved verbatim.

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Allowed standard prerequisites (editorial).

- A morphism is finite if and only if it is integral and locally of finite type. Morphisms, Lemma morphisms-lemma-finite-integral, invoked at proof lines 6325-6327. Splits finiteness into two properties.
- Local finiteness of type and integrality are fpqc local on the base. Same chapter, Lemmas lemma-descending-property-locally-finite-type and lemma-descending-property-integral, invoked at proof lines 6328-6330. Gives both ingredients, hence the result.

Source reference. Original lemma 02L9: sources/stacks-descent/pinned-original-source.tex, source lines 6304–6304. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KX: sources/stacks-descent/pinned-original-source.tex, source lines 5954–5954. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01WJ: sources/stacks-chow/tags.txt, source lines 2484–2484. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is finite" is fpqc local on the base.*

Proof. An finite morphism is the same thing as an integral morphism which is locally of finite type. See Morphisms, Lemma 01WJ. Hence the lemma follows on combining Lemmas 02KX and 02L9. □

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